

Forensic & Disclosure *Defensibility Brief*

Why metadata removal matters in litigation, disclosure and regulatory review — and how PDF Cleanse evidences a defensible process.

DOCUMENT Forensic & Disclosure Brief

COMPANION TO PDF Cleanse Technical
Whitepaper v1.7

VERSION 1.1

PUBLISHED May 2026

AUDIENCE Solicitors, DPOs, FOI officers,
in-house counsel

CLASSIFICATION Public – distribution
permitted

When a document is disclosed externally — in litigation, in response to a subject access request, or under freedom of information legislation — the question that matters is not *whether* metadata was removed. It is *whether* the process by which it was removed is **defensible**: explainable to a regulator, repeatable across a team, and evidenced on the record.

This brief sets out the legal context in which metadata removal arises, describes precisely what PDF Cleanse removes and how, and explains how the audit certificate and processing model together evidence a defensible workflow. It is intended as a companion to the PDF Cleanse Technical Whitepaper, which covers the underlying architecture in greater detail.

§ 01

Why metadata removal matters in disclosure

Three converging pressures make hidden metadata a substantive — not theoretical — risk in any disclosure context.

The legal duty

Under UK GDPR Article 5(1)(c), personal data must be limited to what is necessary for the purpose of processing. Article 17 establishes a right to erasure. When a document is shared externally, any personal data embedded in its metadata layers travels with it — including data that was never intended to be part of the disclosure. This creates a direct exposure: the recipient becomes a third party in receipt of personal data that the discloser had no lawful basis to transmit.

The UK Information Commissioner's Office addresses this directly in its guidance on disclosing documents to the public securely, which warns that hidden personal information in shared files can result in accidental breach when responding to FOI or subject access requests. The Solicitors Regulation Authority's professional conduct rules impose a parallel duty of competence in handling client information.

The litigation context

Two cases illustrate the consequences of inadequate process. In *Hur v. Lloyd & Williams, LLC*, 523 P.3d 861 (Wash. App. Div. 3, 31 January 2023), counsel produced more than 1,000 pages of discovery in which privileged passages were blacked out on the visible page, but the underlying text remained recoverable from the file's metadata layer through a routine keyword search. The court held that a lawyer's duty of competence extends to understanding metadata, and that failure to strip embedded data before disclosure is itself a basis for sanctions. The Italian Garante's Decision No. 243 of 29 April 2025 fined the Lombardy Region €50,000 for unlawful retention of employee email metadata — the first enforcement application of the Garante's 2024 workplace metadata guidelines, and a clear signal that metadata handling is now a regulated activity in its own right.

The procedural test

The standard a regulator or opposing counsel will apply is not "did you try" — it is **"can you demonstrate a defensible, repeatable process?"** A defensible process has four characteristics: it is documented, it is applied consistently, it produces an artefact evidencing each application, and it can be explained in plain language to a non-technical reader.

IN PRACTICE

In a disclosure dispute, the question that decides the outcome is rarely the technical adequacy of the tool. It is whether the disclosing party can produce a record showing that the right step was taken, at the right time, by an identified person, for an identified file. The audit certificate is that record.

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What PDF Cleanse removes

PDF Cleanse applies a four-stage removal process to every file. Each stage addresses a distinct layer of the PDF specification (ISO 32000) and removes a distinct class of identifying or potentially identifying information.

STAGE	PDF SPECIFICATION LAYER	WHAT IS REMOVED
Stage 1 Info dictionary	Document Information Dictionary, defined in ISO 32000-1 §14.3.3	Title, Author, Subject, Keywords, Creator software, Producer software, Creation date, Modification date.
		Adobe's extended metadata — typically a superset of the Info dictionary plus

STAGE	PDF SPECIFICATION LAYER	WHAT IS REMOVED
Stage 2 XMP packets	Extensible Metadata Platform packets, embedded as XML streams (ISO 16684-1)	application-specific fields. Invisible to most PDF viewers but readable by any forensic or scripted tool.
Stage 3 Document IDs	File trailer / ID array (ISO 32000-1 §14.4)	Two hexadecimal fingerprints written at file creation and on each revision. Used to correlate document versions and to identify a file across copies. Overwritten with zero values, preserving structural validity.
Stage 4 PieceInfo streams	Application data dictionaries, defined in ISO 32000-1 §14.5	Workflow data written by publishing applications including Adobe InDesign, Illustrator and Acrobat — internal project structures, operator identifiers, and revision stage information.

What is not removed

PDF Cleanse removes structural metadata. It does not modify the rendered content of a document. Specifically, it does not redact text or images visible on the page, strip EXIF or GPS data from images embedded inside the PDF, or alter embedded fonts. These tasks require separate tools. For high-stakes disclosure, a dedicated redaction step should be applied before or after metadata removal, as appropriate to the document.

§ 03

How removal is performed

The processing model is the single most important element of PDF Cleanse's defensibility posture, because it directly affects the chain of custody analysis.

Client-side processing

PDF Cleanse executes entirely within the user's browser. The file is read into browser memory using the standard FileReader API; processing is performed in memory using the pdf-lib JavaScript library (v1.17.1) and native string operations; the cleaned file is offered for download as a Blob. No network request is made at any point during processing. This is verifiable by any user: browser developer tools (F12 → Network tab) will show zero requests during a cleaning operation.

Implications for chain of custody

Because no file data is transmitted to any server, no third-party processor enters the chain of custody. The file remains under the sole control of the user throughout. This has three direct consequences for disclosure analysis:

- **No data processor relationship** arises between the user's organisation and PDF Cleanse under UK GDPR Article 28. No Data Processing Agreement is required.

- **No Business Associate Agreement** is required under HIPAA, because no protected health information is transmitted to a third party.
- **No additional disclosure exposure** is created by the cleaning step itself, because the document does not leave the disclosing organisation's control.

Per-file operations

Each file is processed through the four stages in sequence: Info dictionary fields are cleared using pdf-lib's setter methods; XMP packets are removed by detecting the standard begin/end markers and excising the intervening bytes; Document IDs are located by regular expression and overwritten with thirty-two zero hex characters; PieceInfo dictionaries are matched and removed. The output is then validated for a correct PDF header before being offered for download. Any file that fails this validation is rejected with an explicit error; the original file is never altered.

§ 04

The audit certificate as evidence

Every cleaning session produces a downloadable plain-text audit certificate. The certificate records, for each file processed:

- File name and file size at the point of processing
- Timestamp in ISO 8601 UTC
- The specific metadata fields that were found and stripped, by name
- All four stages applied, listed explicitly
- **Input SHA-256** — a cryptographic hash of the original file as received
- **Output SHA-256** — a cryptographic hash of the cleaned file as produced
- Result of the operation (Clean, or Error with the underlying message)
- Tool identification and a statement of the processing model
- The standard disclaimer regarding legal advice and the verification requirement

Cryptographic verification

The two SHA-256 values transform the certificate from a self-attested record into an independently verifiable artefact. Any recipient of a cleaned file can compute its SHA-256 hash using a single standard operating-system command and compare it against the Output SHA-256 recorded on the certificate. A matching hash establishes — mathematically, without reliance on the producing party's word or on PDF Cleanse itself — that the file in the recipient's possession is bit-identical to the file the tool recorded as clean. The Input SHA-256 fixes the original file's state at the moment of processing, allowing the same form of verification against the source document if it is later produced.

The cryptographic primitive is the SHA-256 algorithm (FIPS 180-4), computed through the browser's native `crypto.subtle.digest` API. No third-party cryptographic library is involved. Hashes are written into the certificate text and the on-screen confirmation; they are never transmitted, because no transmission occurs.

Evidential weight

The certificate is a contemporaneous record of process. In a disclosure dispute it serves four functions: it evidences that the cleaning step was performed; it identifies the specific fields that were present and therefore removed; it locates the action in time, supporting reconstruction of the disclosure timeline; and through the two SHA-256 values it allows the integrity of both the source and the cleaned file to be verified by any party with access to either. It is suitable for inclusion in DSAR response logs, litigation disclosure records, FOIA processing files and internal data governance audit trails.

RECOMMENDED PAIRING

For high-stakes disclosure (litigation, regulator-facing DSAR, contested FOIA), the audit certificate should be paired with a second-reviewer verification step in Adobe Acrobat. The reviewer opens the cleaned file, navigates to File → Properties, and confirms the metadata fields are clear. A short file-note recording this verification, alongside the certificate, produces a two-step evidenced process that is straightforward to defend.

§ 05

Defensibility summary

If the metadata-handling step in a disclosure is challenged — by opposing counsel in litigation, by the ICO in response to a complaint, or by a data subject in correspondence — the disclosing party can demonstrate the following:

1. **A documented process.** The technical whitepaper describes the four-stage removal architecture and the standards it addresses.
2. **A consistent process.** The tool applies the same four stages to every file, producing reproducible output.
3. **A processing model that does not create new exposure.** Client-side execution means the document does not leave the disclosing organisation.
4. **A contemporaneous record per file.** The audit certificate identifies what was removed, when, and from which file.
5. **Cryptographic integrity.** The SHA-256 hashes of input and output allow any later party to verify that the file in their possession is bit-identical to the file the tool recorded as clean.
6. **An independent verification step.** The cleaned output can be verified in any PDF viewer's Document Properties dialog, providing a confirmation that does not depend on the tool itself.

Together, these elements form a defensible workflow suitable for routine disclosure and adequate for evidencing reasonable steps in regulator-facing or litigation-facing contexts.

Important. This document is a technical and procedural reference. It does not constitute legal advice and does not by itself establish compliance with any law, regulation or professional rule. Organisations remain responsible for their disclosure obligations and should consult their data protection officer or legal counsel

where the disclosure context is contested, regulator-facing or of significant value. PDF Cleanse is a technical utility provided as-is, without warranty as to fitness for any specific disclosure context.